RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Additional Practice Questions for Chapter S2A: Discrete Random Variables

1 John is given 4 keys of which only one can unlock the door. He tries the keys randomly (without replacement) to unlock the door. The random variable X denotes the number of tries John takes to unlock the door.

(i) Show that $\mathrm{P}(X = 3) = \frac{1}{4}$.

(ii) Tabulate the probability distribution of X.

(iii) Write down $\mathrm{E}(X)$ and find $\mathrm{Var}(X)$. Hence find $\mathrm{E}(X + 2)$ and $\mathrm{Var}(X + 2)$.

CJC Prelim 9233/2006/04/Q23(modified)

Solution:

(i)

$$\mathrm{P}\left(X = 3\right) = \mathrm{P}\left(\text{1st two keys are wrong and 3rd key is correct}\right)$$

$$= \frac{3}{4}\cdot\frac{2}{3}\cdot\frac{1}{2} = \frac{1}{4}$$

(ii)

x	1	2	3	4
$\mathrm{P}(X = x)$	$\frac{1}{4}$	$\frac{3}{4}\times\frac{1}{3} = \frac{1}{4}$	$\frac{1}{4}$	$\frac{3}{4}\times\frac{2}{3}\times\frac{1}{2} = \frac{1}{4}$

(iii)

By symmetry, $\mathrm{E}(X) = \frac{2+3}{2} = \frac{5}{2}$

$$\mathrm{E}\left(X^2\right) = \frac{1}{4} + 2^2\left(\frac{1}{4}\right) + 3^2\left(\frac{1}{4}\right) + 4^2\left(\frac{1}{4}\right) = \frac{15}{2}$$

$$\mathrm{Var}(X) = \mathrm{E}\left(X^2\right) - \left[\mathrm{E}\left(X\right)\right]^2 = \frac{15}{2} - \left(\frac{5}{2}\right)^2 = \frac{5}{4}$$

$$\mathrm{E}(X + 2) = \mathrm{E}(X) + 2 = \frac{9}{2}$$

$$\mathrm{Var}\left(X + 2\right) = \mathrm{Var}\left(X\right) = \frac{5}{4}$$

2 A discrete random variable X takes values 2, 3, 4, 5 with probabilities as shown in the table.

x	2	3	4	5
$\mathrm{P}(X=x)$	k	$\frac{k}{4}$	$\frac{k}{9}$	$\frac{k}{16}$

(i) Find k, leaving your answer as a fraction. [2]

(ii) Find $\mathrm{E}\left(\lvert\cos X\rvert\right)$, giving your answer to 3 significant figures. [3]

(iii) Find $\mathrm{P}\left(X_1+X_2=7\right)$, where X_1 and X_2 are two independent observations of X. [3]

MJC Prelim 9233/2005/02/Q12	
Solution:	
(i)	$\sum \mathrm{P}\left(X=x\right)=1$ $k+\frac{k}{4}+\frac{k}{9}+\frac{k}{16}=1$ $k=\frac{144}{205}$ X : denotes the random variable x : denotes the value that the r.v. can take
(ii)	$\mathrm{E}\left(\lvert\cos X\rvert\right)$ Recall: $\mathrm{E}\left[g(X)\right]=\sum g(x)\mathrm{P}\left(X=x\right)$ $=\sum \lvert\cos x\rvert\ \mathrm{P}\left(X=x\right)$ $=\frac{144}{205}\left[\lvert\cos 2\rvert+\frac{\lvert\cos 3\rvert}{4}+\frac{\lvert\cos 4\rvert}{9}+\frac{\lvert\cos 5\rvert}{16}\right]$ $=0.530$ (3 s.f.)
(iii)	$\mathrm{P}\left(X_1+X_2=7\right)=2\ \mathrm{P}\left(X_1=2 \text{ and } X_2=5\right)+2\ \mathrm{P}\left(X_1=3 \text{ and } X_2=4\right)$ $=2k\cdot\frac{k}{16}+2\cdot\frac{k}{4}\cdot\frac{k}{9}$ $=\frac{13}{72}k^2$ $=0.0891$ (3 s.f)

3 A bag contains five red and two black counters.

(a) Mary draws four counters from the bag, one by one at random and without replacement. The random variable X is used to denote the number of red counters drawn by Mary. Show that $\mathrm{P}(X=2)=\dfrac{2}{7}$ and find the expectation of X. [4]

(b) Peter pays \$1 each time to draw a counter at random from the bag, one by one and with replacement until a black counter is obtained. Find the amount of money he should receive when a black counter is drawn in order for the game to be fair. [3]

[You may assume $1+2x+3x^2+4x^3+\ldots=\dfrac{1}{(1-x)^2}$ for $|x|<1$.]

JJC Prelim 9233/2005/02/Q12	
Solution:	
(a)	Let X be the number of red counters drawn. X can take values 2, 3 or 4. $\mathrm{P}(X=2)=\mathrm{P}(2\text{ R }\&\ 2\text{ B in any order})$ $=\dfrac{5}{7}\times\dfrac{4}{6}\times\dfrac{2}{5}\times\dfrac{1}{4}\times\dfrac{4!}{2!2!}=\dfrac{2}{7}$ or $\dfrac{{}^5\mathrm{C}_2\cdot{}^2\mathrm{C}_2}{{}^7\mathrm{C}_4}$ $\mathrm{P}(X=3)=\mathrm{P}(3\text{ R }\&\ 1\text{B in any order})$ $=\dfrac{5}{7}\times\dfrac{4}{6}\times\dfrac{3}{5}\times\dfrac{2}{4}\times\dfrac{4!}{3!}=\dfrac{4}{7}$ or $\dfrac{{}^5\mathrm{C}_3\cdot{}^2\mathrm{C}_1}{{}^7\mathrm{C}_4}$ $\mathrm{P}(X=4)=\mathrm{P}(4\text{ R})$ $=\dfrac{5}{7}\times\dfrac{4}{6}\times\dfrac{3}{5}\times\dfrac{2}{4}=\dfrac{1}{7}$ or $\dfrac{{}^5\mathrm{C}_4\cdot{}^2\mathrm{C}_0}{{}^7\mathrm{C}_4}$ $\mathrm{E}(X)=2\times\dfrac{2}{7}+3\times\dfrac{4}{7}+4\times\dfrac{1}{7}=\dfrac{20}{7}$
(b)	For fair game, E(amt. received) = E(amt. paid) $=\$1\times\dfrac{2}{7}+\$2\times\dfrac{5}{7}\times\dfrac{2}{7}+\$3\times\left(\dfrac{5}{7}\right)^2\times\dfrac{2}{7}+\$4\times\left(\dfrac{5}{7}\right)^3\times\dfrac{2}{7}+\cdots$ $=\$\dfrac{2}{7}[1+2\times\dfrac{5}{7}+3\times\left(\dfrac{5}{7}\right)^2+4\times\left(\dfrac{5}{7}\right)^3+\cdots]$ $=\$\dfrac{2}{7}\cdot\dfrac{1}{(1-\frac{5}{7})^2}$ $=\$3.50$

4 A biased cubical die has faces numbered 1, 2, 3, 4, 5, 6 and X denotes the number obtained when the die is tossed once with probabilities $\mathrm{P}(X = x) = kx^2$. Find the value of k. [2]

Joyce tosses the die and notes the number x obtained. At each toss, she will be given \$($3x$) if $x = 1, 2, 3$, and \$(x) if $x = 4, 5, 6$. Calculate the expected amount Joyce will receive after ten tosses. [4]

NYJC Prelim 9233/2005/02/Q12

Solution:

$$\sum_{x=1}^{6} kx^2 = 1 \quad \Rightarrow \quad 91k = 1 \quad \Rightarrow \quad k = \frac{1}{91}$$

Let \$$T$ be the amount she gets after one toss.

x	1	2	3	4	5	6
t	3	6	9	4	5	6
$\mathrm{P}(T = t)$	$\frac{1^2}{91}$	$\frac{2^2}{91}$	$\frac{3^2}{91}$	$\frac{4^2}{91}$	$\frac{5^2}{91}$	$\frac{6^2}{91}$

$$\mathrm{E}(T) = 3\left(\frac{1}{91}\right) + 6\left(\frac{4}{91} + \frac{36}{91}\right) + 9\left(\frac{9}{91}\right) + 4\left(\frac{16}{91}\right) + 5\left(\frac{25}{91}\right) = \frac{513}{91}$$

Hence expected amount she gets after ten tosses is

$$\mathrm{E}(T_1 + T_2 + \ldots + T_{10}) = 10\mathrm{E}(T) = \frac{5130}{91} = \$56.37 \text{ (nearest cent)}$$

5 X denotes the random variable representing the number of thunderstorms occurring per month. The probability distribution function of X is given in the following table.

x	0	1	2	3	4
$\mathrm{P}(X=x)$	a	0.2	0.5	0.1	b

(i) State the values of a and b which maximizes the value of E(X). [2]

Using these values of a and b,

(ii) Show that $\operatorname{Var} X = \dfrac{101}{100}$. [3]

(iii) Two independent observations X_1 and X_2 are taken of X. Find $\mathrm{P}(X_1 - X_2 \geq 1)$.
Hence, or otherwise, find $\mathrm{P}(|X_1 - X_2| \geq 1)$. [5]

RJC Prelim 9233/2005/04/Q25(modified)

Solution:

(i)

$\sum \mathrm{P}(X=x) = a+0.2+0.5+0.1+b=1 \;\Rightarrow\; a+b=0.2$

$\mathrm{E}(X)=0(a)+1(0.2)+2(0.5)+3(0.1)+4(b)$ is maximum when $a = 0$, $b = 0.2$

(ii)

$\mathrm{E}(X)=1(0.2)+2(0.5)+3(0.1)+4(0.2)=2.3$

$\mathrm{E}(X^2)=1^2(0.2)+2^2(0.5)+3^2(0.1)+4^2(0.2)=6.3$

$\operatorname{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2 = 6.3-(2.3)^2 = \dfrac{101}{100}$

(iii)

$x_1 - x_2$		x_2 = 0	1	2	3	4
x_1	0	0	-1	-2	-3	-4
	1	1	0	-1	-2	-3
	2	2	1	0	-1	-2
	3	3	2	1	0	-1
	4	4	3	2	1	0

$$\mathrm{P}(X_1-X_2\geq 1)=\mathrm{P}(X_1=4,X_2\leq 3)+\mathrm{P}(X_1=3,X_2\leq 2)+\mathrm{P}(X_1=2,X_2\leq 1)+\mathrm{P}(X_1=1,X_2=0)$$
$$=0.2(0.8)+0.1(0.7)+0.5(0.2)+0.2(0)$$
$$=\frac{33}{100}$$

$$\mathrm{P}(|X_1-X_2|\geq 1)=\mathrm{P}(X_1-X_2\geq 1)+\mathrm{P}(X_1-X_2\leq -1)$$
$$=\frac{33}{100}+\frac{33}{100}=\frac{33}{50}$$

6 In a game with a 4-sided fair die numbered 1 to 4 on each face, the score for a throw is the number on the bottom face of the die. A player gets to choose either option A or option B.

Option A: The player rolls the die once. The score x is the amount of money $\$x$ that the player wins.

Option B: The player rolls the die twice. The first score is x and the second score is y. If $y > x$, the player wins $\$2xy$, but if $y < x$, the player loses $\$(x - y)$. Otherwise, he neither wins nor loses any money.

(i) Find the expected amounts won by a player in one game when playing option A and when playing option B. Show that option B is a better option. [5]

(ii) Suggest why a risk averse player would still choose option A. [1]

(iii) Show that the variance of the amount won by a player in one game when playing option A is $\$^2 1.25$. [2]

HCI Prelim 9758/2019/02/Q10 (Part)

Solution:

(i) Let $\$V$ and $\$W$ denote the amounts won by a player in one game when playing options A and B respectively.

$$\mathrm{E}(V) = 1\left(\frac{1}{4}\right) + 2\left(\frac{1}{4}\right) + 3\left(\frac{1}{4}\right) + 4\left(\frac{1}{4}\right)$$

$$= (1+2+3+4)\left(\frac{1}{4}\right) = \frac{5}{2}$$

Hence, expected amount won by a player in one game when playing option A is \$2.50

Under option B, the winning amount for each combination is listed below:

x \ y	1	2	3	4
1	0	4	6	8
2	-1	0	12	16
3	-2	-1	0	24
4	-3	-2	-1	0

$$\mathrm{E}(W) = \frac{1}{16}(4+6+12+8+16+24-1-2-3-1-2-1)$$

$$= \frac{1}{16}(70-10) = \frac{15}{4}$$

Hence, expected amount won by a player in one game when playing option B is \$3.75

Since $\mathrm{E}(W) > \mathrm{E}(V)$, option B is better.

(ii) Option *A* is a "sure win" option where the player would definitely gain a positive amount in all cases, whereas option *B* has a risk of losing money in some cases.

(iii)

$$\mathrm{E}(V^2) = (1^2 + 2^2 + 3^2 + 4^2)\left(\frac{1}{4}\right) = \frac{15}{2}$$

$$\mathrm{Var}(V) = \mathrm{E}(V^2) - [\mathrm{E}(V)]^2 = \frac{15}{2} - \left(\frac{5}{2}\right)^2 = 1.25$$

7 A bag contains 3 white balls and r red balls, where $r \geq 2$. Balls are drawn one at a time, at random from the bag and without replacement. The random variable X is the number of white balls drawn until 2 red balls are drawn.

(i) Show that $\mathrm{P}(X=1)=\dfrac{6r(r-1)}{(r+3)(r+2)(r+1)}$, and find the probability distribution of X. [4]

(ii) Find the expectation of X and show that the variance of X can be expressed in the form $\dfrac{6\mathrm{g}(r)}{(r+2)(r+1)^2}$, where $\mathrm{g}\left(r\right)$ is a quadratic polynomial to be determined. [5]

(iii) Given that the variance of X is $\dfrac{11}{16}$, find the value of r. [1]

TMJC Prelim 9758/2021/02/Q8

Solution:

(i)

$$\mathrm{P}\left(X=0\right)=\mathrm{P}\left(\mathrm{RR}\right)=\frac{r}{r+3}\cdot\frac{r-1}{r+2}$$

$$\mathrm{P}\left(X=1\right)=\mathrm{P}\left(\mathrm{WRR\ or\ RWR}\right)=\frac{3}{r+3}\cdot\frac{r}{r+2}\cdot\frac{r-1}{r+1}\cdot 2=\frac{6r(r-1)}{(r+3)(r+2)(r+1)}$$

$$\mathrm{P}\left(X=2\right)=\mathrm{P}\left(\mathrm{WWRR\ or\ WRWR\ \ or\ RWWR}\right)$$
$$=\frac{3}{r+3}\cdot\frac{2}{r+2}\cdot\frac{r}{r+1}\cdot\frac{r-1}{r}\cdot 3=\frac{18(r-1)}{(r+3)(r+2)(r+1)}$$

$$\mathrm{P}\left(X=3\right)=\mathrm{P}\left(\mathrm{WWWRR\ or\ WWRWR\ or\ WRWWR\ or\ RWWWR}\right)$$
$$=\frac{3}{r+3}\cdot\frac{2}{r+2}\cdot\frac{1}{r+1}\cdot\frac{r}{r}\cdot\frac{r-1}{r-1}\cdot 4=\frac{24}{(r+3)(r+2)(r+1)}$$

x	0	1	2	3
$\mathrm{P}(X=x)$	$\frac{r(r-1)}{(r+3)(r+2)}$	$\frac{6r(r-1)}{(r+3)(r+2)(r+1)}$	$\frac{18(r-1)}{(r+3)(r+2)(r+1)}$	$\frac{24}{(r+3)(r+2)(r+1)}$

(ii)

$$\mathrm{E}(X)=\frac{6r(r-1)}{(r+3)(r+2)(r+1)}+\frac{36(r-1)}{(r+3)(r+2)(r+1)}+\frac{72}{(r+3)(r+2)(r+1)}$$
$$=\frac{6\left(r^2+5r+6\right)}{(r+3)(r+2)(r+1)}=\frac{6\left(r+2\right)(r+3)}{(r+3)(r+2)(r+1)}=\frac{6}{r+1}$$

$$\mathrm{E}(X^2)=\frac{6r(r-1)}{(r+3)(r+2)(r+1)}+\frac{72(r-1)}{(r+3)(r+2)(r+1)}+\frac{216}{(r+3)(r+2)(r+1)}$$
$$=\frac{6(r^2+11r+24)}{(r+3)(r+2)(r+1)}=\frac{6(r+3)(r+8)}{(r+3)(r+2)(r+1)}=\frac{6\left(r+8\right)}{(r+2)(r+1)}$$

	$\mathrm{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2$ $=\dfrac{6(r+8)}{(r+2)(r+1)}-\left(\dfrac{6}{r+1}\right)^2$ $=\dfrac{6(r+8)(r+1)-36(r+2)}{(r+2)(r+1)^2}$ $=\dfrac{6(r^2+3r-4)}{(r+2)(r+1)^2}$
(iii)	$\dfrac{6(r^2+3r-4)}{(r+2)(r+1)^2}=\dfrac{11}{16}$ From GC, $r=7$

8 The probability of obtaining a '6' on a biased cubical dice is thrice the probability of rolling every other number on the dice.

Show that probability of obtaining a '6' on the biased dice is $\frac{3}{8}$. [1]

This biased dice is put into a bag together with 3 fair cubical dice.

(i) One of the dice is chosen randomly from the bag and rolled once. Find the probability of obtaining a '6'. [2]

(ii) One of the dice is chosen randomly and is rolled n times.

(a) Find the probability that '6' is obtained on all the n rolls. [1]

(b) Given that '6' is obtained for all the n rolls, the probability that the biased dice is chosen is more than 0.95. By forming an inequality in terms of n, solve for the least value of n. [3]

(iii) One of the dice is chosen randomly from the bag and rolled once. The dice is then put back into the bag. Another dice is chosen randomly from the bag and rolled once.

For every '6' obtained, the player gains \$2, otherwise the player loses \$$k$. If the game is fair, that is, expected winnings of the player is \$0, show that the value of k is 0.56. [3]

Find the variance of the player's winnings. [2]

VJC Prelim 9758/2018/02/Q8

Solution:

Let the probability of obtaining a '6' be k .

$$k+5\cdot\frac{1}{3}k=1$$

$$\frac{8}{3}k=1 \quad\Rightarrow\quad k=\frac{3}{8}$$

(i) P(obtaining a '6') = $\frac{1}{4}\times\frac{3}{8}+\frac{3}{4}\times\frac{1}{6}=\frac{7}{32}$

(ii) (a) P(obtaining n '6's) = $\frac{1}{4}\times\left(\frac{3}{8}\right)^n+\frac{3}{4}\times\left(\frac{1}{6}\right)^n$

(ii) (b)

$$\frac{\frac{1}{4}\times\left(\frac{3}{8}\right)^n}{\frac{1}{4}\times\left(\frac{3}{8}\right)^n+\frac{3}{4}\times\left(\frac{1}{6}\right)^n}>0.95$$

n	Probability
4	0.89521 < 0.95
5	0.95055 > 0.95

Least n is 5.

(iii) Let X (in \$) be the winnings from randomly choosing one dice and rolling it once.
The winnings of the player who chooses 2 dice with replacement and rolls them is $X_1 + X_2$.
Given that the expected winnings is \$0,

$$\mathrm{E}(X_1 + X_2) = 0$$
$$2\mathrm{E}(X) = 0$$
$$\mathrm{E}(X) = 0$$

x	$-k$	2
$\mathrm{P}(X = x)$	$\frac{25}{32}$	$\frac{7}{32}$

$$\mathrm{E}(X) = 0$$
$$\frac{25}{32}(-k) + \frac{7}{32}(2) = 0$$
$$k = 0.56$$

x	-0.56	2
$\mathrm{P}(X = x)$	$\frac{25}{32}$	$\frac{7}{32}$

From GC, $\mathrm{Var}(X) = 1.12$

$\mathrm{Var}(X_1 + X_2) = 2\,\mathrm{Var}(X) = 2.24$

The variance of the player's winnings is $\$^2 2.24$.

OR

Let \$$Y$ be the winnings of the player who chooses 2 dice with replacement and rolls them.

y	$-2k$	$2-k$	4
$\mathrm{P}(Y = \mathrm{y})$	$\left(\frac{25}{32}\right)^2 = \frac{625}{1024}$	$2\left(\frac{7}{32}\right)\left(\frac{25}{32}\right) = \frac{175}{512}$	$\left(\frac{7}{32}\right)^2 = \frac{49}{1024}$

$$\mathrm{E}(Y) = -2k\left(\frac{625}{1024}\right) + (2-k)\left(\frac{175}{512}\right) + 4\left(\frac{49}{1024}\right)$$
$$0 = -\frac{625k}{512} + \frac{175}{256} - \frac{175k}{512} + \frac{49}{256}$$
$$\frac{25}{16}k = \frac{7}{8}$$
$$k = 0.56$$

y	-1.12	1.44	4
$\mathrm{P}(Y = y)$	$\frac{625}{1024}$	$\frac{175}{512}$	$\frac{49}{1024}$

From GC, $\mathrm{Var}(Y) = 2.24$

The variance of the player's winnings is $\$^2 2.24$.

9 Shania and Tina are playing a game. Shania has a bag containing one green disc, r red discs and $2r$ blue discs, where $r > 1$. A red disc is worth 5 points, a blue disc is worth 2 points and the green disc is worth 0 points. Tina takes two discs from the bag at random. Tina's score is found by multiplying together the number of points for each of the two discs she takes.

(i) State Tina's possible scores. [1]

(ii) Show that the expectation of Tina's score is $\dfrac{27r-11}{3r+1}$, and find an expression for the variance. [7]

(iii) Given that the variance of Tina's score is 38, find the value of r. [2]

9758/2020/02/Q5		
Solution:		
(i)	0, 4, 10, 25	
(ii)	$\mathrm{P}(S=0)=\dfrac{{}^{1}\mathrm{C}_1\,{}^{3r}\mathrm{C}_1}{{}^{3r+1}\mathrm{C}_2}$ or $\dfrac{1}{3r+1}\cdot\dfrac{3r}{3r}\cdot 2=\dfrac{2}{3r+1}$ $\mathrm{P}(S=4)=\dfrac{{}^{2r}\mathrm{C}_2}{{}^{3r+1}\mathrm{C}_2}$ or $\dfrac{2r}{3r+1}\cdot\dfrac{2r-1}{3r}=\dfrac{2(2r-1)}{3(3r+1)}$ $\mathrm{P}(S=10)=\dfrac{{}^{r}\mathrm{C}_1\,{}^{2r}\mathrm{C}_1}{{}^{3r+1}\mathrm{C}_2}$ or $\dfrac{r}{3r+1}\cdot\dfrac{2r}{3r}\cdot 2=\dfrac{4r}{3(3r+1)}$ $\mathrm{P}(S=25)=\dfrac{{}^{r}\mathrm{C}_2}{{}^{3r+1}\mathrm{C}_2}$ or $\dfrac{r}{3r+1}\cdot\dfrac{r-1}{3r}=\dfrac{r-1}{3(3r+1)}$	Good to check sum of prob =1 $\dfrac{2}{3r+1}+\dfrac{\frac{2}{3}(2r-1)}{3r+1}+\dfrac{\frac{4}{3}r}{3r+1}+\dfrac{\frac{1}{3}(r-1)}{3r+1}$ $=\dfrac{1}{3r+1}\left(2+\dfrac{4}{3}r-\dfrac{2}{3}+\dfrac{4}{3}r+\dfrac{1}{3}r-\dfrac{1}{3}\right)$ $=1$
	$\mathrm{E}(S)=\dfrac{1}{3r+1}\left(0\times 2+4\times\dfrac{2}{3}(2r-1)+10\times\dfrac{4}{3}r+25\times\dfrac{1}{3}(r-1)\right)$ $=\dfrac{1}{3r+1}\left(\dfrac{16}{3}r-\dfrac{8}{3}+\dfrac{40}{3}r+\dfrac{25}{3}r-\dfrac{25}{3}\right)=\dfrac{1}{3r+1}(27r-11)$ $\mathrm{E}(S^2)=\dfrac{1}{3r+1}\left(0^2\times 2+4^2\times\dfrac{2}{3}(2r-1)+10^2\times\dfrac{4}{3}r+25^2\times\dfrac{1}{3}(r-1)\right)$ $=\dfrac{1}{3r+1}\left(\dfrac{64}{3}r-\dfrac{32}{3}+\dfrac{400}{3}r+\dfrac{625}{3}r-\dfrac{625}{3}\right)=\dfrac{1}{3r+1}(363r-219)$ $\mathrm{Var}(S)=\mathrm{E}(S^2)-[\mathrm{E}(S)]^2$ $=\dfrac{1}{3r+1}(363r-219)-\left[\dfrac{1}{3r+1}(27r-11)\right]^2$ $=\dfrac{1}{(3r+1)^2}\left[(363r-219)(3r+1)-(27r-11)^2\right]=\dfrac{1}{(3r+1)^2}\left[360r^2+300r-340\right]$	
(iii)	$\dfrac{1}{(3r+1)^2}\left[360r^2+300r-340\right]=38$ By GC, $r=3$	

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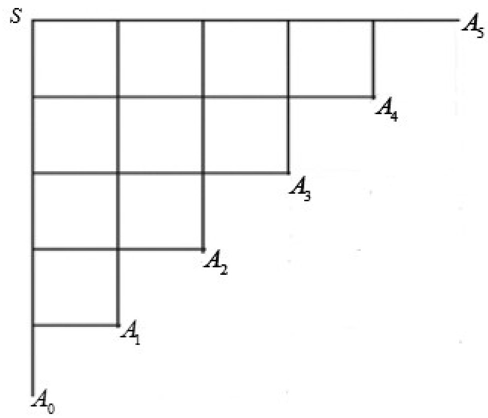

A particle moves one step each time either to the right or downwards through a network of connected paths as shown. The particle starts at S, and, at each junction, randomly moves one step to the right with probability p, or one step downwards with probability q, where $q = 1 - p$. The steps taken at each junction are independent. The particle finishes its journey at one of the 6 points labelled A_i, where $i = 0, 1, 2, 3, 4, 5$ (see diagram). Let $\{ X = i \}$ be the event that the particle arrives at point A_i.

(i) Show that $\mathrm{P}(X = 2) = 10p^2q^3$. [2]

(ii) After experimenting, it is found that the particle will end up at point A_2 most of the time. By considering the mode of X or otherwise, show that $\frac{1}{3} < p < \frac{1}{2}$. [4]

The above setup is a part of a two-stage computer game.

- If the particle lands on A_0, the game ends immediately and the player will not win any points.
- If the particle lands on A_i, where $i = 2$ or 4, then the player gains 2 points.
- If the particle lands on A_i, where $i = 1, 3$ or 5, then the player proceeds on to the next stage, where there is a probability of 0.4 of winning the stage. If he wins the stage, he gains 5 points; otherwise he gains 3 points.

Let Y be the number of points gained by the player when one game is played.

(iii) If $p = 0.4$, determine the probability distribution of Y. [4]

(iv) Hence find the expectation and variance of Y. [2]

MI Prelim 9758/2019/02/Q10

Solution:

(i)

$\mathrm{P}(X=2)=\mathrm{P}(\text{point } A_2)$

$=\mathrm{P}(2 \text{ steps to the right and 3 steps downwards in any order})$

$={}^5\mathrm{C}_2\, p^2q^3=10p^2q^3$

(ii)

Since the particle will end up at point A_2 most of the time, then mode occurs at $X=2$.

$\mathrm{P}(X=1)<\mathrm{P}(X=2)>\mathrm{P}(X=3)$

Note that $\mathrm{P}(X=1)=\mathrm{P}(\text{point } A_1)={}^5\mathrm{C}_1\, pq^4=5pq^4$,

and $\mathrm{P}(X=3)=\mathrm{P}(\text{point } A_3)={}^5\mathrm{C}_3\, p^3q^2=10p^3q^2$

Hence $5pq^4<10p^2q^3$ and $10p^2q^3>10p^3q^2$

$q<2p$ since $p,\ q>0$ $\qquad$ $q>p$ since $p,\ q>0$

$1-p<2p$ and $1-p>p$

$p>\dfrac{1}{3}$ $\qquad$ $p<\dfrac{1}{2}$

Combining both inequalities, $\dfrac{1}{3}<p<\dfrac{1}{2}$

(iii)

$\mathrm{P}(Y=0)=\mathrm{P}(X=0)={}^5\mathrm{C}_0(0.4)^0(0.6)^5=0.07776$

$\mathrm{P}(Y=2)=\mathrm{P}(X=2)+\mathrm{P}(X=4)$

$=10(0.4)^2(0.6)^3+{}^5\mathrm{C}_4(0.4)^4(0.6)^1=0.4224$

$\mathrm{P}(Y=3)=\mathrm{P}(X=i, \text{ where } i \text{ is odd and lose the 2nd stage})$

$=(\mathrm{P}(X=1)+\mathrm{P}(X=3)+\mathrm{P}(X=5))\times 0.6$

$=\left[{}^5\mathrm{C}_1(0.4)^1(0.6)^4+{}^5\mathrm{C}_3(0.4)^3(0.6)^2+{}^5\mathrm{C}_5(0.4)^5(0.6)^0\right]\times 0.6$

$=0.299904$

$\mathrm{P}(Y=5)=\mathrm{P}(X=i, \text{ where } i \text{ is odd and wins the 2nd stage})$

$=(\mathrm{P}(X=1)+\mathrm{P}(X=3)+\mathrm{P}(X=5))\times 0.4$

$=\left[{}^5\mathrm{C}_1(0.4)^1(0.6)^4+{}^5\mathrm{C}_3(0.4)^3(0.6)^2+{}^5\mathrm{C}_5(0.4)^5(0.6)^0\right]\times 0.4$

$=0.199936$

The probability distribution of Y is:

y	0	2	3	5
$\mathrm{P}(Y=y)$	0.07776	0.4224	0.299904	0.199936

(iv)

Using G.C., $\mathrm{E}(Y)=2.74$ (3 sf) and $\mathrm{Var}(Y)=1.3626^2=1.86$ (3 sf)

THE END